

OFFICIAL
ACCIDENT CASE
ANALYSIS //
DOCUMENTARY
RECORD

THE MANGALORE DISASTER

*Technical Investigation into Flight Path
Dynamics, Human Factors, and Table-Top
Airfield Limitations*



Accident Airframe: Boeing 737-8HG (VT-AXV)

Airframe Profile:	Boeing 737-800 Next Generation (VT-AXV)
Flight Vector:	IX-812 // Scheduled Dubai (DXB) to Mangalore
Incident Envelope:	(IXE) May 22, 2010 // 06:05 Hours Indian Standard Time
Investigation Basis:	Court of Inquiry presided by Air Marshal B.N. Gokhale

1. OPERATIONAL PROFILE & CASUALTY METRICS

On the morning of May 22, 2010, Air India Express Flight IX-812 approached its terminal vector at Mangalore International Airport after a routine international transit from Dubai [cite: 4885, 4890]. The aircraft overshot the paved surface of Runway 24, cleared the primary boundary safety margins, tore through the Instrument Landing System (ILS) ground localizer antenna housing, and plunged into a deep wooded gorge [cite: 4904, 4911, 4912]. Instantly upon hull fragmentation, a violent fuel-fed thermal emergency occurred, completely incinerating the airframe structure [cite: 4913, 5048].

The aircraft carried 160 passengers (including 4 infants) and 6 active crew members [cite: 4886, 4891]. In the catastrophic impact sequence and subsequent fire, 152 passengers and all 6 crew members suffered fatal deceleration and thermal injuries [cite: 4913]. Only 8 passengers survived, having been thrown clear or managed to escape through breaks in the cabin walls before the structure was totally engulfed by toxic smoke and flames [cite: 4913, 5597, 5600].

1.1 Flight Deck Crew Profile

The operational management of the aircraft was allocated to two primary crew members [cite: 5117]:

Crew Allocation	Pilot-in-Command (PIC) [cite: 4855]	First Officer (FO) [cite: 4855]
Identity	Captain Zlatko Glusica [cite: 4855]	Co-pilot Harbinder Singh Ahluwalia [cite: 5075]
Nationality / Age	Serbian / 55 Years [cite: 5051, 5072]	Indian / 40 Years [cite: 5076, 5095]
Total Logged Hours	10,215:50 Hours [cite: 5073]	3,620:00 Hours [cite: 5096]
Logged Hours on Type	2,844:50 Hours (Boeing 737-800) [cite: 5073]	3,319:40 Hours (Boeing 737-800) [cite: 5096]
Airport Route Experience	16 Landings at Mangalore Airport [cite: 4981]	66 Sectors flown into Mangalore Airport [cite: 4981]

1.2 Localized Environmental Data at Timeline of Impact

The terminal weather recorded by the Aeronautical Meteorological Station at Mangalore indicated the following operational parameters at the time of arrival (0030 UTC / 05:30 IST baseline updated through landing clearance at 06:04 IST) [cite: 5369, 5371]:

Environmental Parameter	Official Metric Value [cite: 5371]	Operational Impact Framework [cite: 5010, 5458]
Surface Wind Speed	Calm [cite: 5371]	Zero crosswind or strong headwind component detected [cite: 5010].
Horizontal Visibility	6 Kilometers [cite: 5371]	Visual approach environment established; morning haze observed [cite: 5357, 5371].
Atmospheric Cloud Profile	Few Clouds at 2,000 ft [cite: 5371]	Stable ceiling base over the tabletop plateau [cite: 5371].
Temperature / Dew Point	27°C / 26°C [cite: 5371]	High relative ambient air moisture levels [cite: 5371].
Altimeter Setting (QNH)	1006 hPa [cite: 5371]	Standard local pressure parameters verified across altimeters [cite: 5371].
Precipitation / Runway State	None during landing roll // Dry [cite: 5458]	Runway surface clean and completely free of contamination [cite: 5458].

2. AIRCRAFT TECHNICAL OVERVIEW

The airframe, registered as VT-AXV, was a Boeing 737-8HG Next Generation model manufactured in 2008 under Manufacturer's Serial Number 36333 [cite: 5112]. Prior to the accident flight, the airframe had logged 7,199:41 hours across 2,833 landing cycles [cite: 5124, 5125]. The plane was equipped with twin high-bypass CFM56-7B27/3 turbofan powerplants, producing 27,300 pounds of thrust each [cite: 5111]. To navigate complex technical topography like Mangalore's short fields, the aircraft was configured with Boeing's factory Short Field Performance (SFP) design adjustments [cite: 7140].

A rigorous review of the airframe logbooks, technical maintenance documents, and post-accident system teardowns verified that the plane was in a state of continuous airworthiness [cite: 6228]. All applicable Airworthiness Directives (ADs) and DGCA mandatory notifications were complied with [cite: 5123]. Only two minor snags were active under the Minimum Equipment List (MEL): an unserviceable passenger seat back on 25C and a localized failure of the Right-Hand Tail Logo Light [cite: 4889, 5167, 5168]. Teardown reports of the brake stacks, stabilizer trim jackscrews, and throttle resolver angles confirmed that the airframe responded directly to all mechanical and fly-by-wire control inputs throughout the entire landing roll sequence [cite: 5637, 5645, 5700].

3. INFRASTRUCTURE PROFILE & TABLE-TOP CONSTRAINTS

Mangalore International Airport presents significant geographic challenges due to its 'Table-Top' layout [cite: 5944]. Leveled across an undulating plateau ridge, Runway 24/06 is bordered on both flanks and its overshoot boundaries by sheer vertical cliffs dropping directly into wooded valleys [cite: 5356, 5413, 5941].



Figure 3.1: Table-Top aerial profile of Runway 24 indicating plateau margins and extreme gorge boundaries.

Runway 24 had a total paved length of 2,450 meters and a width of 46 meters [cite: 5404]. However, due to severe surrounding drops, the flat runway strip extended only 75 meters on either side of the centerline instead of the standard 150 meters [cite: 5405, 5412]. The Runway End Safety Area (RESA) spanned 90 meters by 90 meters, which was the international minimum standard in 2010 [cite: 5406, 5414]. Crucially, the terrain immediately past the RESA descended at a harsh slope toward the airport boundary fence, and a concrete, non-frangible Instrument Landing System (ILS) antenna platform was positioned directly in line with the overshoot path [cite: 5021, 5416, 5417].

4. SWISS CHEESE MODEL ACCIDENT ANALYSIS

To chart the systematic factors behind the crash, investigators utilized James Reason's Swiss Cheese Model, analyzing how latent organizational issues, systemic infrastructure gaps, and active pilot actions aligned to allow the catastrophe:

1. Organizational Influences (Latent Failure Layer): Air India Express functioned as a rapidly growing low-cost entity while drawing senior management Post Holders from its parent carrier [cite: 5751, 5756]. These decision-

makers were not type-rated or operational on the Boeing 737 fleet [cite: 5759]. Flight deck crew coordination was further challenged by a lack of digital automated crew scheduling [cite: 5903]. Rostering was managed manually with pencils and paper, which failed to adequately account for cumulative fatigue patterns during complex multi-base night flights [cite: 5903, 5907, 5908].

2. Environmental and Technological Deficiencies (Latent Layer): On May 20th, the primary rotary joint of the Mangalore Area Surveillance Radar (MSSR) suffered a mechanical failure [cite: 4894, 5385, 5386]. Due to a lack of immediate local replacement parts, the radar remained out of service during the flight's arrival window [cite: 5387, 5388]. Air Traffic Control had to use Procedural Control separation, which limited situational visibility and forced them to clear Flight IX-812 for descent at a tight 77 DME mark instead of the standard 130 DME profile [cite: 4897, 4919, 4974, 4977].

3. Physiological Preconditions (Active Fatigue Layer): Due to the late-night timing of the return flight from Dubai, both crew members faced severe biological pressure [cite: 4945, 4946]. The CVR recorded the Captain sleeping for 1 hour and 40 minutes during the cruise phase [cite: 4895, 4971]. Because he woke up just minutes before the descent into the Window of Circadian Low (WOCL), the Captain suffered from severe *Sleep Inertia*—a period of cognitive impairment and delayed visual processing that severely degraded his ability to accurately assess a high-altitude glide path [cite: 4916, 4917, 4918].

4. Unsafe Acts & Cockpit Resource Breakdown (Active Layer): Because the descent profile was compressed by the tight procedural descent window, the aircraft arrived over the final approach fix at double the standard altitude (crossing the threshold at 200 feet at 164 knots instead of 50 feet at 144 knots) [cite: 4900, 5013]. The Captain experienced cognitive tunnel vision, manually pulling the aircraft down at an excessive sink rate of over 3,300 feet per minute, which triggered several automated EGPWS 'SINK RATE' and 'PULL UP' alerts [cite: 4902, 5006, 5117]. The First Officer repeatedly called to 'Go Around' three times but, due to a steep Trans-Cockpit Authority Gradient, failed to physically intervene or override the Captain's controls until it was too late [cite: 4901, 4921, 5004, 5016].

5. INVESTIGATION THEORIES: CONFIRMATION & ELIMINATION

The Court of Inquiry thoroughly tested several technical and mechanical theories before finalizing the causal lineage of the overrun [cite: 4822, 5623]:

- **Airframe Control or Deceleration Stack Failure (ELIMINATED):** Initial concerns focused on a potential failure of the aircraft's braking or spoiler systems. However, physical inspection of the wreckage and DFDR data confirmed that all four carbon brake stacks operated within Component Maintenance Manual (CMM) limits [cite: 5026, 5704, 5705]. Manual brake application pressure spiked normally to over 2,400 PSI, and the auto-brake system was correctly selected to Position 2, ruling out mechanical deceleration failure [cite: 5019, 5637, 5720].

- **Mid-Air Sabotage or Explosive Event (ELIMINATED):** Given the sudden nature of the impact sequence, teams from the Bureau of Civil Aviation Security (BCAS) and the Bomb Disposal Squad carefully screened all wreckage sections [cite: 5037]. The analysis certified that there were absolutely no chemical traces, structural tearing patterns, or soot marks pointing to an explosive event or sabotage [cite: 4914, 5038, 6293].

- **Avian Engine Ingestion (ELIMINATED):** Engineers stripped down both CFM56 engines at the Air India facility in Mumbai to verify if a bird strike had caused a sudden loss of thrust [cite: 5042, 5674]. The internal compressor blades showed clean mechanical damage consistent with high-energy ground rotation, with no biological residues or organic material detected, completely ruling out a bird strike [cite: 4914, 5703, 5739, 5740].
- **Sleep Inertia and Delayed Deceleration Decisions (CONFIRMED):** The synchronized data from the CVR and DFDR proved that the aircraft was mechanically sound [cite: 4914, 6235]. Boeing flight simulation modeling demonstrated that even with the late touchdown at 5,200 feet, the aircraft could have stopped safely within the remaining 2,800 feet of hard surface if maximum manual braking and reverse thrust were held continuously [cite: 4904, 7129, 7130]. Instead, the Captain made a critical error by stowing the thrust reversers and attempting a 'Go-Around' six seconds after touchdown [cite: 4910, 5019]. This delayed decision made the overrun inevitable, driving the aircraft off the plateau and into the concrete localizer platform [cite: 4911, 4912].

6. POST-ACCIDENT REFORMS & SAFETY EVOLUTION

The definitive report on Flight IX-812 led to significant safety updates across Indian and international commercial aviation regulations [cite: 4922]:

1. **Overhaul of Flight Duty Time Limitations (FDTL):** Archaic FDTL mandates from 1992 were fully replaced by the DGCA [cite: 4922, 5966]. The updated rules enforced longer rest windows between routes, strictly regulated back-to-back night duties, and mandated the implementation of predictive Fatigue Risk Management Systems (FRMS) to address the biological risks of WOCL flight tracking [cite: 4922, 5967, 6779].
2. **Standardization of Stabilized Approach Parameters:** Airlines hard-coded strict 'Stabilized Approach' rules into their standard operating procedures (SOPs) [cite: 5808, 6128]. Under these mandates, if an airframe does not meet specific speed, altitude, and glide slope metrics by 1,000 feet in instrument conditions (or 500 feet in visual conditions), an immediate missed approach is compulsory [cite: 5809, 7161, 7166]. Additionally, 'No-Fault Go-Around' rules protect pilots from any corporate tracking or operational penalties when electing to abort a landing [cite: 6799, 6800].
3. **Civil Infrastructure Upgrades:** The Ministry of Civil Aviation mandated that all domestic airfields expand their Runway End Safety Areas (RESA) from the previous 90-meter baseline up to a safer 240-meter length [cite: 6718, 6721]. Furthermore, any hard concrete ground equipment or structures within airfield boundaries were replaced with fully frangible, low-mass items designed to break apart immediately upon impact [cite: 6714, 6715].

Reported By:
Sujana
Batch 7 Student - Aviation